Customer Churn Analysis

The analysis of the customer churn dataset indicates that 26.54% of customers have discontinued the service, while 73.46% of customers have been retained. This shows that although the majority of customers remain loyal, more than 1 out of every 4 customers leaves the company, making churn reduction an important business objective.

- Customer Demographics

The analysis of gender-wise churn reveals that churn behavior is relatively similar across male and female customers, indicating that gender is not a significant factor influencing customer churn. However, when examining senior citizens, a noticeably higher percentage of customers in the senior citizen category have churned compared to non-senior citizens. This suggests that older customers may face service-related, pricing, or support challenges that increase their likelihood of leaving.

- Customer Tenure Analysis

Customer tenure emerged as one of the strongest indicators of churn. Customers who have been associated with the company for only 1–2 months show the highest churn rates, whereas customers with longer tenures are significantly more likely to stay. This highlights that the initial months of a customer's journey are the most critical period for retention. Effective onboarding, customer support, and engagement strategies during this phase can substantially reduce churn.

- Contract Type Analysis

Contract type has a major impact on customer retention. Customers subscribed to month-to-month contracts exhibit the highest churn percentage, whereas customers with one-year and two-year contracts demonstrate much stronger loyalty and lower churn rates. Long-term contractual commitments appear to strengthen customer retention by increasing engagement and reducing switching behavior.

- Internet Service and Additional Features

The analysis of internet services reveals that customers using Fiber Optic Internet Service have the highest churn rate compared to DSL users and customers without internet service. This could indicate potential concerns related to pricing, service quality, or customer expectations.

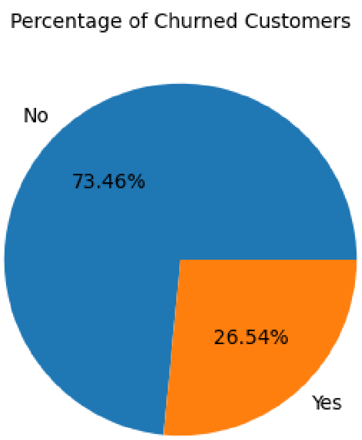
Additionally, customers who do not subscribe to Online Security, Online Backup, Device Protection, and Tech Support services are considerably more likely to churn. These value-added services appear to improve customer confidence and satisfaction, ultimately contributing to better retention.

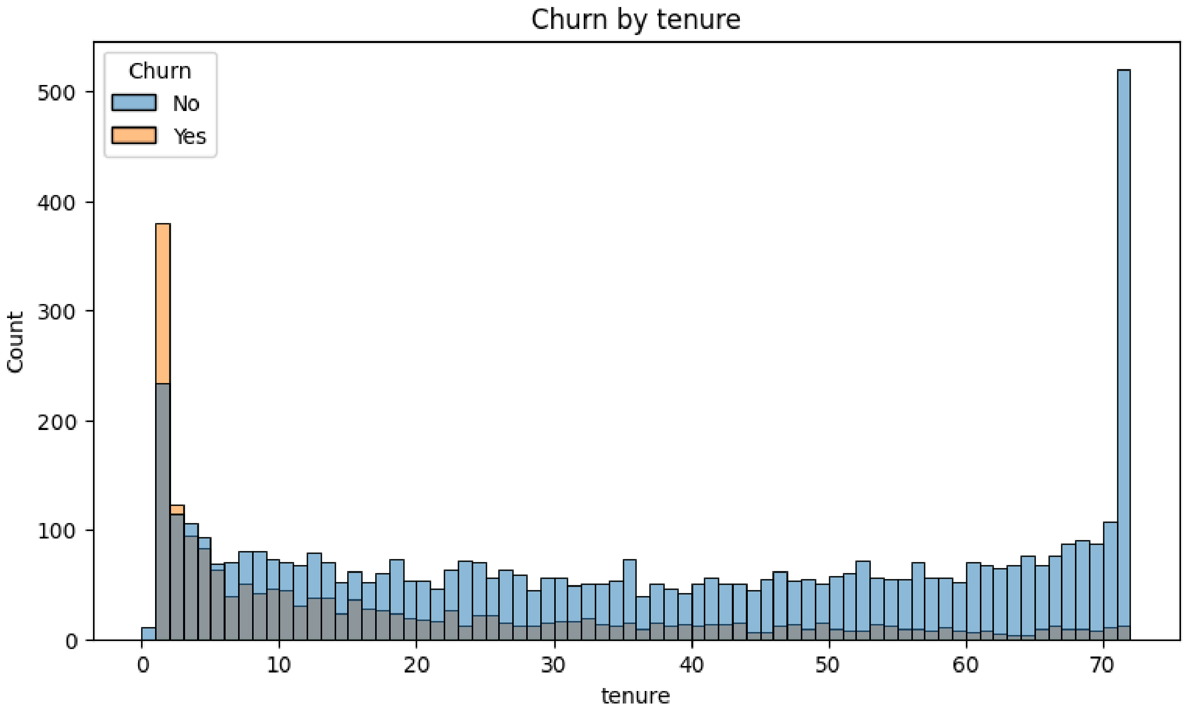
- Payment Method Analysis

Payment method analysis highlights a clear pattern: customers using Electronic Check as their payment method have the highest churn percentage among all payment options. Customers using automated payment methods such as bank transfers or credit cards appear more stable and less likely to leave. This suggests that payment convenience and customer commitment may influence retention behavior.

Key Business Insights

- 26.54% of customers have churned, while 73.46% remain with the company.
- Senior citizens exhibit a higher churn tendency than non-senior customers.
- Customers with 1–2 months tenure are the most vulnerable to churn.
- Month-to-month contract holders represent the highest-risk customer segment.
- Fiber Optic customers show greater churn compared to other internet service categories.
- Customers lacking security, backup, protection, and support services are more likely to leave.
- Electronic Check users demonstrate the highest churn rates.
- Long-term customers and customers on annual contracts show the strongest retention.





Business Recommendations

1. Focus retention campaigns on customers during their first 3 months of service.
2. Encourage migration from month-to-month contracts to long-term plans through discounts and loyalty benefits.
3. Investigate customer satisfaction among Fiber Optic subscribers and address service concerns.
4. Promote Online Security, Backup, Device Protection, and Tech Support bundles to increase customer stickiness.
5. Target Electronic Check users with incentives to switch to automated payment methods.
6. Design specialized retention programs for senior citizens through personalized support and simplified service plans.

Conclusion

The analysis demonstrates that churn is primarily driven by short customer tenure, flexible month-to-month contracts, Fiber Optic service usage, lack of support/security features, and Electronic Check payments. By addressing these high-risk segments and strengthening customer engagement during the early stages of the customer lifecycle, the company can significantly improve retention rates and reduce overall churn.